

CBF-Constrained NMPC as the Expert Planner

Formulation, baseline comparison, and multi-modal scoring for LLM trajectory distillation

Problem formulation

The candidate expert solves a constrained trajectory optimization over a horizon of N steps, coupling a safety certificate with an explicit optimality objective — rather than sampling geometry (RRT) or filtering a reference one step at a time (CBF-QP).

Safety certificate. A control barrier function h defines a safe set that is rendered forward-invariant:

$$\mathcal{C} = \{x : h(x) \geq 0\}, \quad \dot{h}(x, u) \geq -\alpha(h(x)) \quad (1)$$

Joint trajectory optimization. States and controls are solved for *together*, subject to dynamics and safety at every stage:

$$\min_{x_{0:N}, u_{0:N-1}} J(x_{0:N}, u_{0:N-1}) \quad \text{s.t.} \quad x_{k+1} = x_k + \Delta t(f(x_k) + g(x_k)u_k), \quad h(x_k) \geq 0 \quad \forall k \quad (2)$$

Why it matters for the pipeline. Whatever this solve returns becomes (x_d, u_d) in the calibration chain:

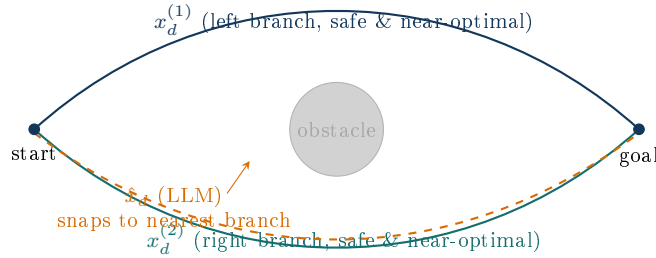
$$\|x - x^\star\| \leq \underbrace{\|x - \hat{x}_d\|}_{\text{contraction}} + \underbrace{\|\hat{x}_d - x_d\|}_{\text{conformal (CP)}} + \underbrace{\|x_d - x^\star\|}_{\rightarrow 0 \text{ here}} \quad (3)$$

Because $h(x_k) \geq 0$ holds by construction along the solution of (2), the third term collapses to (near) zero — the payoff that justifies the offline cost. It also removes the fit-then-differentiate step (waypoints $\rightarrow$ min-snap $\rightarrow$ flatness) that was amplifying $u_d - \hat{u}_d$: x_d, u_d are dynamically consistent by construction, not inferred after the fact.

A nonconvex NLP, solved offline — exactly the property this comparison is built around.

Baseline comparison

Method	Data-gen cost	Safety	Optimality	Fit with $(x_d, \hat{x}_d, u_d, \hat{u}_d)$ formulation
RRT / RRT* (<i>current</i>)	Low — fast sampling	None; asymptotic completeness only	Asymptotic, no margin	Simple, but needs a separate min-snap fit $\Rightarrow$ derivative amplification
Tube / Robust MPC	High — one NMPC solve per traj.	Strong — explicit tube radius	Local, per-horizon	Margin = term 3 directly; x, u still fit post-hoc
SOS Funnels / LQR-Trees	Very high — offline bilinear SOS	Strong — Lyapunov sublevel set	Not the target (stabilization/coverage)	Excellent — composes directly with contraction $M(x)$
HJ Reachability	Extreme — curse of dimensionality	Exact, zero conservatism	N/A — safety-only formulation	Best possible margin; intractable at full 12-D state
CBF-QP (<i>single-step</i>)	Very low — already real-time (ms)	Local invariance; myopic $\Rightarrow$ can go infeasible later	Reference-tracking only, not global	Poor fit — already cheap online, nothing to distill
MIQP Corridor Selection	Very high — NP-hard, combinatorial	Strong — exact corridor containment	Global (branch & bound), if solved to completion	Good margin; discrete choice complicates smooth analysis
CBF/HOCBF + NMPC (<i>horizon, §1</i>)	High — SQP/NLP per solve, genuinely offline	Strong — invariance over horizon + sampled-data care	Strong — explicit horizon-wide J	Best fit — joint x, u consistency removes flatness amplification



Handling multiplicity. A nonconvex NMPC of this form has no unique solution: different initial guesses, or different homotopy choices around an obstacle (left vs. right), can converge to several distinct, individually-safe, individually-near-optimal trajectories $x_d^{(1)}, \dots, x_d^{(M)}$ for the same start-goal-obstacle scene. Rather than forcing the LLM to match one arbitrarily-selected member of that family, the conformity score can instead measure distance to the *nearest* member, $d(\hat{x}_d) = \min_{i \leq M} \|\hat{x}_d - x_d^{(i)}\|$, so a prediction is penalized only for leaving the entire safe family — not for choosing a different-but-equally-valid branch than the one the solver happened to return. This also reframes part of the earlier control-deviation problem: some of that gap may have been mode disagreement rather than a genuine quality gap — the LLM picking an equally-safe alternative the single stored expert trajectory never represented. The cost of this relaxation: each member of the family needs its *own* certificate for the $\rightarrow 0$ term in Eq. (3) to still hold, so M cannot be padded with heuristically-generated near-misses.